



# PostgreSQL

## POSTGRES DATABASE ADMINISTRATION



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|SENIOR DATABASE ADMINISTRATOR|ORACLE| POSTGRESQL | MSSQL|OCI|AWS|

## Chapter 2: PostgreSQL Installation and Initialization

### About the Author:

I am **Stephen Njoroge**, a Senior Database Engineer with over 8+ years of expertise in designing, securing and optimizing high-performance database systems across the banking and fintech sectors. My core technical proficiency spans major databases: Oracle, PostgreSQL, MS SQL Server and MySQL-along with advanced scripting in SQL and Python for automation and data pipeline development. I architect and manage cloud and virtualization solutions on OCI, AWS, and Proxmox for onprem systems and implement comprehensive DevOps practices using monitoring tools like Prometheus and Grafana and IT service management via iTop. Further, I deliver advanced data analytics and visualization using Tableau, Power BI and Oracle BI. As a Certified Data Protection Officer, I integrate robust data protection, compliance and security measures including vulnerability assessment and role-based access control into every layer of the data infrastructure.

# PostgreSQL Available Versions and Supported OS Platforms

PostgreSQL follows a versioning system where major versions (e.g., 15, 16, 17) include significant new features. Each major version is supported for 5 years and runs on multiple platforms including Linux, Windows, macOS, FreeBSD, and Solaris. For Ubuntu, packages are available through the official PostgreSQL repository.

PostgreSQL can be downloaded from [www.postgresql.org/downloads](http://www.postgresql.org/downloads).

## PostgreSQL Available Versions

| PostgreSQL Major Version Release Timeline |                      |                          |                |
|-------------------------------------------|----------------------|--------------------------|----------------|
| PostgreSQL Version                        | Initial Release Date | Final Release / EOL Date | Support Status |
| PostgreSQL 18                             | 25-Sep-25            | 14-Nov-30                | Supported      |
| PostgreSQL 17                             | 26-Sep-24            | 8-Nov-29                 | Supported      |
| PostgreSQL 16                             | 14-Sep-23            | 9-Nov-28                 | Supported      |
| PostgreSQL 15                             | 13-Oct-22            | 11-Nov-27                | Supported      |
| PostgreSQL 14                             | 30-Sep-21            | 12-Nov-26                | Supported      |
| PostgreSQL 13                             | 24-Sep-20            | 13-Nov-25                | End-of-Life    |
| PostgreSQL 12                             | 3-Oct-19             | 21-Nov-24                | End-of-Life    |
| PostgreSQL 11                             | 18-Oct-18            | 9-Nov-23                 | End-of-Life    |
| PostgreSQL 10                             | 5-Oct-17             | 10-Nov-22                | End-of-Life    |

## PostgreSQL supported platforms

| OS Family      | Supported Distributions / Variants                                                                                       |
|----------------|--------------------------------------------------------------------------------------------------------------------------|
| Linux          | <b>Major Distros:</b> Red Hat, Rocky, AlmaLinux, Fedora, Ubuntu, Debian, SUSE. <b>Others:</b> Most modern distributions. |
| Windows        | Desktop (e.g., 2025, 2022, 2019) & Server editions. Also supports Cygwin/MSYS2.                                          |
| macOS (Darwin) | Recent versions (e.g., macOS 13+ for PostgreSQL 18).                                                                     |
| BSD Family     | FreeBSD, OpenBSD, NetBSD, DragonFlyBSD.                                                                                  |
| Solaris Family | Solaris, illumos.                                                                                                        |

# System Requirements for PostgreSQL Installation on Linux

## Minimum System Requirements

### 1. Hardware Requirements

- **CPU:** Modern 64-bit x86, ARM, or compatible processor
- **RAM:** Minimum **1GB** (2GB+ recommended for production)
- **Disk Space:** Approximately **150MB** for binaries, plus space for your databases

- Production: **20GB+** recommended, depending on data size
- **Disk Type:** SSD strongly recommended for production workloads

## 2. Operating System Support

### Officially Supported:

- **RHEL/CentOS/Rocky Linux/AlmaLinux** 7.x, 8.x, 9.x
- **Ubuntu** 20.04, 22.04, 24.04 LTS
- **Debian** 10, 11, 12
- **Fedora** 36+
- **openSUSE** Leap 15.x
- **SUSE Linux Enterprise Server** 15.x
- **Amazon Linux** 2

## 3. Software Dependencies

### Common build/run dependencies:

- **GNU C Library (glibc)**
- **GCC** (if compiling from source)
- **GNU Make** (if compiling from source)
- **Readline** library (readline-devel or libreadline-dev)
- **zlib** (zlib-devel or zlib1g-dev)
- **Systemd** (for modern distributions with systemd integration)
- **OpenSSL** (openssl-devel or libssl-dev) - for encrypted connections
- **libxml2** - for XML support
- **ICU** - for internationalization support
- **UUID libraries** (uuid-devel or uuid-dev)

## 4. Memory Requirements by Use Case

- **Development/Testing:** 2-4 GB RAM
- **Small Production:** 4-8 GB RAM
- **Medium Production:** 8-16 GB RAM
- **Large Production:** 16 GB+ RAM

## 5. File System Requirements

- **Recommended:** **XFS**, **ext4**, or **ZFS**
- **Avoid:** Network file systems for production data directories
- **tmpfs** can be used for temporary tablespaces (if RAM is sufficient)

## 6. Network Requirements

- **Default Port:** 5432 (TCP)
- **Firewall:** Ensure port 5432 is open if allowing remote connections
- **Network bandwidth:** Sufficient for your replication/connection needs

## 7. User/Privilege Requirements

- **Non-root user:** PostgreSQL runs as postgres user (created during installation)
- **Superuser access:** Required for installation
- **SELinux/AppArmor:** May need configuration adjustments

An automated script to check system requirements for installation of PostgreSQL in Linux.

```
#!/bin/bash

# PostgreSQL System Requirements Check Script
# Version: 2.0
# Description: Checks system compatibility for PostgreSQL installation

# Color codes for output
RED='\033[0;31m'
GREEN='\033[0;32m'
YELLOW='\033[1;33m'
BLUE='\033[0;34m'
NC='\033[0m' # No Color
BOLD='\033[1m'

# Minimum requirements
MIN_RAM_MB=1024      # 1 GB
MIN_DISK_MB=500      # 500 MB for binaries
MIN_SWAP_MB=1024     # 1 GB swap (recommended)
PG_PORT=5432
PG_VERSION="16"      # Change to desired version

# Results tracking
PASS_COUNT=0
WARN_COUNT=0
FAIL_COUNT=0

# Log file
LOG_FILE="postgresql_system_check_$(date +%Y%m%d_%H%M%S).log"

# Function to print section headers
print_header() {
    echo -e "\n${BLUE}${BOLD}=== $1 ===${NC}"
    echo -e "${BLUE}${BOLD}=== $1 ===${NC}" >> "$LOG_FILE"
}

# Function to print results
print_result() {
    local status=$1
    local message=$2

    case $status in
        "PASS")
            echo -e "${GREEN}[✓] $message${NC}"
            echo "[PASS] $message" >> "$LOG_FILE"
            ((PASS_COUNT++))
            ;;
        "WARN")
```

```

        echo -e "${YELLOW}[!] $message${NC}"
        echo "[WARN] $message" >> "$LOG_FILE"
        ((WARN_COUNT++))
        ;;
    "FAIL")
        echo -e "${RED}[X] $message${NC}"
        echo "[FAIL] $message" >> "$LOG_FILE"
        ((FAIL_COUNT++))
        ;;
    "INFO")
        echo -e "${BLUE}[i] $message${NC}"
        echo "[INFO] $message" >> "$LOG_FILE"
        ;;
esac
}

# Function to check if command exists
command_exists() {
    command -v "$1" >/dev/null 2>&1
}

# Start logging
echo "PostgreSQL System Requirements Check - $(date)" > "$LOG_FILE"
echo "===== " >> "$LOG_FILE"

print_header "SYSTEM INFORMATION"

# Check OS and version
if [ -f /etc/os-release ]; then
    . /etc/os-release
    OS_NAME="$NAME"
    OS_VERSION="$VERSION_ID"
    OS_ID="$ID"
    print_result "INFO" "Operating System: $OS_NAME $OS_VERSION ($OS_ID)"
else
    OS_NAME=$(uname -s)
    OS_VERSION=$(uname -r)
    print_result "INFO" "Operating System: $OS_NAME $OS_VERSION"
fi

# Check kernel
KERNEL=$(uname -r)
print_result "INFO" "Kernel Version: $KERNEL"

# Check architecture
ARCH=$(uname -m)
if [[ "$ARCH" == "x86_64" ]] || [[ "$ARCH" == "aarch64" ]]; then
    print_result "PASS" "Architecture: $ARCH (Supported)"
else
    print_result "WARN" "Architecture: $ARCH (May require special compilation)"
fi

print_header "HARDWARE REQUIREMENTS"

# Check CPU
CPU_CORES=$(nproc 2>/dev/null || grep -c ^processor /proc/cpuinfo)
print_result "INFO" "CPU Cores: $CPU_CORES"

CPU_INFO=$(lscpu 2>/dev/null | grep "Model name" | cut -d: -f2 | xargs)
if [ -n "$CPU_INFO" ]; then

```

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```

    print_result "INFO" "CPU Model: $CPU_INFO"
fi

# Check RAM
TOTAL_RAM_KB=$(grep MemTotal /proc/meminfo | awk '{print $2}')
TOTAL_RAM_MB=$((TOTAL_RAM_KB / 1024))
TOTAL_RAM_GB=$((TOTAL_RAM_MB / 1024))

if [ $TOTAL_RAM_MB -ge $MIN_RAM_MB ]; then
    print_result "PASS" "RAM: ${TOTAL_RAM_GB}GB (Minimum: $((MIN_RAM_MB/1024))GB)"
else
    print_result "FAIL" "RAM: ${TOTAL_RAM_MB}MB (Minimum: ${MIN_RAM_MB}MB required)"
fi

# Check Swap
TOTAL_SWAP_KB=$(grep SwapTotal /proc/meminfo | awk '{print $2}')
if [ -n "$TOTAL_SWAP_KB" ] && [ "$TOTAL_SWAP_KB" != "0" ]; then
    TOTAL_SWAP_MB=$((TOTAL_SWAP_KB / 1024))
    if [ $TOTAL_SWAP_MB -ge $MIN_SWAP_MB ]; then
        print_result "PASS" "Swap: ${TOTAL_SWAP_MB}MB (Minimum: ${MIN_SWAP_MB}MB)"
    else
        print_result "WARN" "Swap: ${TOTAL_SWAP_MB}MB (Recommended: ${MIN_SWAP_MB}MB)"
    fi
else
    print_result "WARN" "No swap space configured"
fi

print_header "STORAGE REQUIREMENTS"

# Check disk space for root partition
ROOT_DISK_KB=$(df -k / | tail -1 | awk '{print $4}')
ROOT_DISK_MB=$((ROOT_DISK_KB / 1024))
ROOT_DISK_GB=$((ROOT_DISK_MB / 1024))

if [ $ROOT_DISK_MB -ge $MIN_DISK_MB ]; then
    print_result "PASS" "Root Disk Space: ${ROOT_DISK_GB}GB (Minimum: $((MIN_DISK_MB/1024))GB)"
else
    print_result "FAIL" "Root Disk Space: ${ROOT_DISK_MB}MB (Minimum: ${MIN_DISK_MB}MB)"
fi

# Check /var partition (common location for PostgreSQL data)
if mount | grep -q " /var "; then
    VAR_DISK_KB=$(df -k /var | tail -1 | awk '{print $4}')
    VAR_DISK_MB=$((VAR_DISK_KB / 1024))
    VAR_DISK_GB=$((VAR_DISK_MB / 1024))
    print_result "INFO" "/var Partition Space: ${VAR_DISK_GB}GB"
fi

# Check filesystem type
FS_TYPE=$(df -T / | tail -1 | awk '{print $2}')
RECOMMENDED_FS=("xfs" "ext4" "zfs" "btrfs")
if [[ " ${RECOMMENDED_FS[@]} " =~ " ${FS_TYPE} " ]]; then
    print_result "PASS" "Filesystem: $FS_TYPE (Recommended)"
else
    print_result "WARN" "Filesystem: $FS_TYPE (Recommended: XFS, ext4, ZFS)"
fi

print_header "SOFTWARE DEPENDENCIES"

# Check package manager
declare -A PKG_MANAGERS=(

```

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```

["apt"]="apt-get"
["yum"]="yum"
["dnf"]="dnf"
["zypper"]="zypper"
["pacman"]="pacman"
)

PKG_MANAGER=""
for pm in "${!PKG_MANAGERS[@]}"; do
    if command_exists "$pm"; then
        PKG_MANAGER="${PKG_MANAGERS[$pm]}"
        break
    fi
done

if [ -n "$PKG_MANAGER" ]; then
    print_result "PASS" "Package Manager: $PKG_MANAGER"
else
    print_result "WARN" "No standard package manager found"
fi

# Check essential dependencies
ESSENTIAL_DEPS=("gcc" "make" "readline" "zlib")
MISSING_DEPS=()

print_result "INFO" "Checking essential build dependencies..."

case $OS_ID in
    "ubuntu"|"debian")
        for dep in "${ESSENTIAL_DEPS[@]}"; do
            if dpkg -l | grep -q "$dep"; then
                print_result "PASS" "Package: $dep (installed)"
            else
                print_result "WARN" "Package: $dep (not installed)"
                MISSING_DEPS+=("$dep")
            fi
        done
        ;;
    "rhel"|"centos"|"fedora"|"rocky"|"almalinux")
        for dep in "${ESSENTIAL_DEPS[@]}"; do
            if rpm -qa | grep -q "$dep"; then
                print_result "PASS" "Package: $dep (installed)"
            else
                print_result "WARN" "Package: $dep (not installed)"
                MISSING_DEPS+=("$dep")
            fi
        done
        ;;
    *)
        print_result "INFO" "Skipping package checks for $OS_ID"
        ;;
esac

# Check optional dependencies
OPTIONAL_DEPS=("openssl" "libxml2" "icu")
print_result "INFO" "Checking optional dependencies..."

for dep in "${OPTIONAL_DEPS[@]}"; do
    case $OS_ID in
        "ubuntu"|"debian")
            if dpkg -l | grep -q "$dep"; then

```

```

        print_result "PASS" "Optional: $dep (installed)"
    else
        print_result "INFO" "Optional: $dep (not installed - optional)"
    fi
    ;;
"rhel"|"centos"|"fedora"|"rocky"|"almalinux")
    if rpm -qa | grep -q "$dep"; then
        print_result "PASS" "Optional: $dep (installed)"
    else
        print_result "INFO" "Optional: $dep (not installed - optional)"
    fi
    ;;
esac
done

print_header "NETWORK CONFIGURATION"

# Check if PostgreSQL port is in use
if command_exists "netstat"; then
    if netstat -tuln | grep -q ":${PG_PORT} "; then
        print_result "FAIL" "Port $PG_PORT is already in use"
    else
        print_result "PASS" "Port $PG_PORT is available"
    fi
elif command_exists "ss"; then
    if ss -tuln | grep -q ":${PG_PORT} "; then
        print_result "FAIL" "Port $PG_PORT is already in use"
    else
        print_result "PASS" "Port $PG_PORT is available"
    fi
else
    print_result "WARN" "Cannot check port $PG_PORT (netstat/ss not found)"
fi

# Check network interfaces
print_result "INFO" "Network Interfaces:"
ip -br addr show | head -5 | while read -r line; do
    print_result "INFO" "  $line"
done

print_header "EXISTING POSTGRESQL INSTALLATIONS"

# Check if PostgreSQL is already installed
if command_exists "psql"; then
    PG_VERSION_INSTALLED=$(psql --version | awk '{print $3}')
    print_result "WARN" "PostgreSQL $PG_VERSION_INSTALLED is already installed"
else
    print_result "PASS" "No existing PostgreSQL installation found"
fi

# Check for PostgreSQL service
if systemctl list-unit-files | grep -q postgresql; then
    print_result "WARN" "PostgreSQL service unit exists"
fi

print_header "SECURITY CONTEXT"

# Check SELinux status
if command_exists "sestatus"; then
    SELINUX_STATUS=$(sestatus | grep "SELinux status" | awk '{print $3}')
    if [ "$SELINUX_STATUS" == "enabled" ]; then

```

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```

        print_result "INFO" "SELinux: Enabled (may require configuration)"
    else
        print_result "INFO" "SELinux: Disabled"
    fi
fi

# Check AppArmor status
if command_exists "aa-status"; then
    if aa-status --enabled 2>/dev/null; then
        print_result "INFO" "AppArmor: Enabled (may require configuration)"
    fi
fi

print_header "USER AND PERMISSIONS"

# Check current user
CURRENT_USER=$(whoami)
print_result "INFO" "Current user: $CURRENT_USER"

# Check sudo access
if sudo -n true 2>/dev/null; then
    print_result "PASS" "User has sudo privileges"
else
    print_result "WARN" "User may not have sudo privileges"
fi

# Check if postgres user exists
if getent passwd postgres > /dev/null; then
    print_result "INFO" "User 'postgres' already exists"
else
    print_result "INFO" "User 'postgres' will be created during installation"
fi

print_header "RECOMMENDATIONS"

# Generate recommendations based on findings
echo -e "${BOLD}Installation Recommendations:${NC}"
echo "-----"

if [[ "$OS_ID" == "ubuntu" ]] || [[ "$OS_ID" == "debian" ]]; then
    echo "1. Install using: sudo apt install postgresql-$PG_VERSION postgresql-contrib-$PG_VERSION"
    if [ ${#MISSING_DEPS[@]} -gt 0 ]; then
        echo "2. Install missing dependencies: sudo apt install ${MISSING_DEPS[*]}"
    fi
elif [[ "$OS_ID" == "rhel" ]] || [[ "$OS_ID" == "centos" ]] || [[ "$OS_ID" == "rocky" ]] || [[ "$OS_ID" == "almalinux" ]]; then
    echo "1. Enable PostgreSQL repository first:"
    echo "    sudo dnf install -y https://download.postgresql.org/pub/repos/yum/reporpms/EL-$
(rpm -E %rhel)-x86_64/pgdg-redhat-repo-latest.noarch.rpm"
    echo "2. Install PostgreSQL: sudo dnf install -y postgresql$PG_VERSION-server postgresql$PG_VERSION-contrib"
elif [[ "$OS_ID" == "fedora" ]]; then
    echo "1. Install: sudo dnf install postgresql-server postgresql-contrib"
fi

if [ $TOTAL_RAM_MB -lt 2048 ]; then
    echo "3. Consider increasing RAM to at least 2GB for better performance"
fi

if [ $ROOT_DISK_MB -lt 10240 ]; then

```

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```

    echo "4. Ensure sufficient disk space (10GB+ recommended for production)"
fi

print_header "SUMMARY"

# Display summary
echo -e "${BOLD}Check Results:${NC}"
echo "-----"
echo -e "${GREEN}Passed:  $PASS_COUNT${NC}"
echo -e "${YELLOW}Warnings: $WARN_COUNT${NC}"
echo -e "${RED}Failed:   $FAIL_COUNT${NC}"
echo "-----"

if [ $FAIL_COUNT -eq 0 ]; then
    if [ $WARN_COUNT -eq 0 ]; then
        echo -e "${GREEN}${BOLD}System is fully compatible for PostgreSQL installation!${NC}"
        EXIT_CODE=0
    else
        echo -e "${YELLOW}${BOLD}System is compatible with some warnings. Review
recommendations.${NC}"
        EXIT_CODE=1
    fi
else
    echo -e "${RED}${BOLD}System has compatibility issues. Fix failures before
installation.${NC}"
    EXIT_CODE=2
fi

echo -e "\n${BLUE}Detailed log saved to: $LOG_FILE${NC}"
echo -e "${BLUE}Run 'cat $LOG_FILE' to review full results${NC}"

exit $EXIT_CODE

```

*save and run as : postgresql\_system\_check.sh*

*./ postgresql\_system\_check.sh*

```
root@998-986-02:/home/steve/scripts# ./postgresql_system_check.sh
```

#### === SYSTEM INFORMATION ===

```
[i] Operating System: Ubuntu 24.04 (ubuntu)
[i] Kernel Version: 5.15.167.4-microsoft-standard-WSL2
[✓] Architecture: x86_64 (Supported)
```

#### === HARDWARE REQUIREMENTS ===

```
[i] CPU Cores: 8
[i] CPU Model: Intel(R) Core(TM) i7-10510U CPU @ 1.80GHz
[✓] RAM: 3GB (Minimum: 1GB)
[✓] Swap: 1024MB (Minimum: 1024MB)
```

#### === STORAGE REQUIREMENTS ===

```
[✓] Root Disk Space: 953GB (Minimum: 0GB)
[✓] Filesystem: ext4 (Recommended)
```

#### === SOFTWARE DEPENDENCIES ===

```
[✓] Package Manager: apt-get
[i] Checking essential build dependencies...
[✓] Package: gcc (installed)
[✓] Package: make (installed)
[✓] Package: readline (installed)
[✓] Package: zlib (installed)
[i] Checking optional dependencies...
[✓] Optional: openssl (installed)
[✓] Optional: libxml2 (installed)
[✓] Optional: icu (installed)
```

#### === NETWORK CONFIGURATION ===

```
[X] Port 5432 is already in use
[i] Network Interfaces:
[i]   lo                UNKNOWN      127.0.0.1/8 10.255.255.254/32 ::1/128
[i]   eth0              UP           192.168.248.237/20 fe80::215:5dff:fe53:9adf/64
```

#### === EXISTING POSTGRESQL INSTALLATIONS ===

```
[!] PostgreSQL 16.11 is already installed
[!] PostgreSQL service unit exists
```

#### === SECURITY CONTEXT ===

#### === USER AND PERMISSIONS ===

```
[i] Current user: root
[✓] User has sudo privileges
[i] User 'postgres' already exists
```

#### === RECOMMENDATIONS ===

Installation Recommendations:

```
1. Install using: sudo apt install postgresql-16 postgresql-contrib-16
```

#### === SUMMARY ===

Check Results:

```
Passed: 14
Warnings: 2
Failed: 1
```

## Environment Details

*Setting up the environment with appropriate hostname and preparing for PostgreSQL installation.*

*Update hostname if not set.*

```
hostnamectl set-hostname Training_svr
```

```
root@998-986-02:/# hostnamectl set-hostname Training_svr
root@998-986-02:/# hostname
Trainingsvr
root@998-986-02:/# |
```

```
hostnamectl set-hostname Training_svr && \
echo "Training_svr" > /etc/hostname && \
sed -i "s/127.0.1.1.*/127.0.1.1\tTraining_svr/" /etc/hosts && \
export PS1='\[\e[1;32m\]\u@Training_svr\[\e[0m\]:\[\e[1;34m\]\w\[\e[0m\]# '
```

```
root@998-986-02:/# hostnamectl set-hostname Training_svr && \
echo "Training_svr" > /etc/hostname && \
sed -i "s/127.0.1.1.*/127.0.1.1\tTraining_svr/" /etc/hosts && \
export PS1='\[\e[1;32m\]\u@Training_svr\[\e[0m\]:\[\e[1;34m\]\w\[\e[0m\]# '
root@Training_svr:/#
```

Verify hostname

```
root@Training_Svr:/# hostname
```

```
root@Training_svr:/# hostname
Trainingsvr
root@Training_svr:/#
```

Update package lists

```
root@Training_Svr:/# apt update
```

```
root@Training_svr:/# apt update
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
118 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@Training_svr:/#
```

# PostgreSQL Installation on Ubuntu using Official Repository

## System Update

*Update package list and upgrade existing packages*

```
# apt update
```

```
root@Training_svr:/# apt update
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
2 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@Training_svr:/# |
```

```
# apt upgrade -y
```

```
root@Training_svr:/# apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

Install some basic utilities

```
# apt install -y wget ca-certificates software-properties-common
```

```
root@Training_svr:/# apt install -y wget ca-certificates software-properties-common
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
wget is already the newest version (1.21.4-1ubuntu4.1).
wget set to manually installed.
ca-certificates is already the newest version (20240203).
ca-certificates set to manually installed.
software-properties-common is already the newest version (0.99.49.3).
software-properties-common set to manually installed.
The following packages were automatically installed and are no longer required:
  libdrm-nouveau2 libdrm-radeon1 libgl1-amber-dri libglapi-mesa libxcb-dri2-0
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.
root@Training_svr:/# |
```

# Add PostgreSQL Official Repository

## Import PostgreSQL GPG Key

*Download and import the repository signing key*

```
sudo wget -qO- https://www.postgresql.org/media/keys/ACCC4CF8.asc | sudo tee  
/etc/apt/trusted.gpg.d/postgresql.asc
```

```
root@Training_svr:/# wget -qO- https://www.postgresql.org/media/keys/ACCC4CF8.asc | tee /etc/apt/trusted.gpg.d/postgresql.asc  
-----BEGIN PGP PUBLIC KEY BLOCK-----  
  
mQINBE6XR8IBEACVdDKT2HEH1IyHzXkb4nIWAY7echjRxo7MTcj4vbXAYBK0fjja  
UrBEJWHN6fjKJXOYWXHLIYg0h0GeW9qcSiaa1/rYIb0zjfGfhE4x0Y+NJHS1db0V  
G6GUj3qXaeyqIJGS2z7m0Thy4Lgr/LpZLZ78Nf1fLiSzBLMo1sV7PpP/7zU0+aA4  
bKa8Rio3weMXQ0ZgcLzgeSdqtWkNyKTQdXY5MkH1QXyFIk1nTfWwyqpJjHlgtwMi  
c2cxjqG5nnV9rIYLTtjYG6RBglq0SmzF/raBnF4Lwjxq4qRqvRLlBXdFu5+2pMfC  
IZ10HPRdQDCTN60DUix+BTzBUT30NzaLhZbOMT5RvQtvTVgWpeIn20i2NrPWNCUh  
hj490dKDLpK/v+A5/i8zPvN4c6MkDH1lFZfaoz3863dyLUBR3Ip26oM0hHXf4/2U  
A/oA4pCL2W0hc4aNTozjKHKvjRx5Q8/hVYu+39csFWxo6YSB/KgIEw+0W8DiTII3  
RQj/OlD68ZDmGLyQPiJvaEtY9fDrcSpI0Esm0i4sjkNbuuh0Cvwwwqo5EF1zfKvj  
Tqz2REYQGMJGc5LubIpk5sMHo1HWV038TWxLDRwt0dzw08zQA6BeWe9F0okRPeR2  
AqhyaJJwOZJodKZ76S+LDwFkTLzEKnYPCzkoRwLrEdNt1M7wQBThnC5z6wARAQAB  
-----END PGP PUBLIC KEY BLOCK-----
```

## Add PostgreSQL Repository

*First, check your Ubuntu version*

```
lsb_release -cs
```

```
root@Training_svr:/# lsb_release -cs  
No LSB modules are available.  
noble  
root@Training_svr:/# cat /etc/os-release  
PRETTY_NAME="Ubuntu 24.04.3 LTS"  
NAME="Ubuntu"  
VERSION_ID="24.04"  
VERSION="24.04.3 LTS (Noble Numbat)"  
VERSION_CODENAME=noble  
ID=ubuntu  
ID_LIKE=debian  
HOME_URL="https://www.ubuntu.com/"  
SUPPORT_URL="https://help.ubuntu.com/"  
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"  
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"  
UBUNTU_CODENAME=noble  
LOGO=ubuntu-logo  
root@Training_svr:/#
```

Add repository

```
sh -c 'echo "deb [arch=amd64] http://apt.postgresql.org/pub/repos/apt $(lsb_release -cs)-pgdg main" >  
/etc/apt/sources.list.d/pgdg.list'
```

## Chapter 2: PostgreSQL Installation and Initialization

# Update Package List Again

Update package list to include PostgreSQL repository

```
# apt update
```

```
root@Training_svr:/# apt update
Get:1 http://apt.postgresql.org/pub/repos/apt noble-pgdg InRelease [107 kB]
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Get:6 http://apt.postgresql.org/pub/repos/apt noble-pgdg/main amd64 Packages [353 kB]
Fetched 460 kB in 2s (256 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
9 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@Training_svr:/#
```

```
root@Training_svr:/# apt list --upgradable
Listing... Done
libgl1-amber-dri/noble-updates 21.3.9-0ubuntu3~24.04.1 amd64 [upgradable from: 21.3.9-0ubuntu2]
libglapi-mesa/noble-updates 24.2.8-1ubuntu1~24.04.1 amd64 [upgradable from: 24.0.9-0ubuntu0.1]
libpq5/noble-pgdg 18.1-1.pgdg24.04+2 amd64 [upgradable from: 16.11-0ubuntu0.24.04.1]
postgresql-16/noble-pgdg 16.11-1.pgdg24.04+1 amd64 [upgradable from: 16.11-0ubuntu0.24.04.1]
postgresql-client-16/noble-pgdg 16.11-1.pgdg24.04+1 amd64 [upgradable from: 16.11-0ubuntu0.24.04.1]
postgresql-client-common/noble-pgdg 287.pgdg24.04+1 all [upgradable from: 257build1.1]
postgresql-common/noble-pgdg 287.pgdg24.04+1 all [upgradable from: 257build1.1]
postgresql/noble-pgdg 18+287.pgdg24.04+1 all [upgradable from: 16+257build1.1]
python3-psycopg2/noble-pgdg 2.9.10-1.pgdg24.04+1 amd64 [upgradable from: 2.9.9-1build1]
root@Training_svr:/# |
```

```
# apt upgrade -y
```

```
root@Training_svr:/# apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  libdrm-nouveau2 libdrm-radeon1 libgl1-amber-dri libglapi-mesa libllvm17t64 libxcb-dri2-0
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  libllvm19 libnuma1 liburing2 postgresql-18 postgresql-18-jit postgresql-client-18
The following packages have been kept back:
  libgl1-amber-dri libglapi-mesa
The following packages will be upgraded:
  libpq5 postgresql postgresql-16 postgresql-client-16 postgresql-client-common postgresql-common python3-psycopg2
7 upgraded, 6 newly installed, 0 to remove and 2 not upgraded.
Need to get 67.1 MB of archives.
After this operation, 218 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 libnuma1 amd64 2.0.18-1build1 [23.3 kB]
```

```

Disabling automatic startup of old cluster...
Starting upgraded cluster on port 5432...
Running finish phase upgrade hook scripts ...
vacuumdb: processing database "postgres": Generating minimal optimizer statistics (1 target)
vacuumdb: processing database "template1": Generating minimal optimizer statistics (1 target)
vacuumdb: processing database "training": Generating minimal optimizer statistics (1 target)
vacuumdb: processing database "postgres": Generating medium optimizer statistics (10 targets)
vacuumdb: processing database "template1": Generating medium optimizer statistics (10 targets)
vacuumdb: processing database "training": Generating medium optimizer statistics (10 targets)
vacuumdb: processing database "postgres": Generating default (full) optimizer statistics
vacuumdb: processing database "template1": Generating default (full) optimizer statistics
vacuumdb: processing database "training": Generating default (full) optimizer statistics
vacuumdb: vacuuming database "postgres"
vacuumdb: vacuuming database "template1"
vacuumdb: vacuuming database "training"

```

Success. Please check that the upgraded cluster works. If it does,  
you can remove the old cluster with

```
pg_dropcluster 16 main
```

```

Ver Cluster Port Status Owner   Data directory           Log file
16  main    5433 down   postgres /var/lib/postgresql/16/main log/postgresql-%Y-%m-%d_%H%M%S.csv.csv
Ver Cluster Port Status Owner   Data directory           Log file
18  main    5432 online postgres /var/lib/postgresql/18/main log/postgresql-%Y-%m-%d_%H%M%S.csv.csv
Setting up postgresql-16 (16.11-1.pgdg24.04+1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...

```

## Check Available PostgreSQL Versions

List available PostgreSQL versions

```
apt-cache search postgresql | grep ^postgresql-
```

```

postgresql-server-dev-18 - development files for PostgreSQL 18 server-side programming
postgresql-server-dev-18-dbg - debug symbols for postgresql-server-dev-18
postgresql-server-dev-8.2 - development files for PostgreSQL 8.2 server-side programming
postgresql-server-dev-8.3 - development files for PostgreSQL 8.3 server-side programming
postgresql-server-dev-8.4 - development files for PostgreSQL 8.4 server-side programming
postgresql-server-dev-9.0 - development files for PostgreSQL 9.0 server-side programming
postgresql-server-dev-9.1 - development files for PostgreSQL 9.1 server-side programming
postgresql-server-dev-9.2 - development files for PostgreSQL 9.2 server-side programming
postgresql-server-dev-9.3 - development files for PostgreSQL 9.3 server-side programming
postgresql-server-dev-9.4 - development files for PostgreSQL 9.4 server-side programming
postgresql-server-dev-9.5 - development files for PostgreSQL 9.5 server-side programming
postgresql-server-dev-9.6 - development files for PostgreSQL 9.6 server-side programming
postgresql-server-dev-all - build tools for multiple PostgreSQL versions
postgresql-contrib - additional facilities for PostgreSQL (supported version)
postgresql-autodoc - Utility to create a PostgreSQL database schema overview in HTML, DOT and XML
postgresql-comparator - efficient PostgreSQL table content comparison and synchronization
root@Training_svr:/#

```

## Install PostgreSQL

### Install Latest Version

*Install PostgreSQL server and client*

```
apt install -y postgresql postgresql-contrib
```

### Chapter 2: PostgreSQL Installation and Initialization

```

root@Training_svr:/# apt install -y postgresql postgresql-contrib
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
postgresql is already the newest version (18+287.pgdg24.04+1).
postgresql-contrib is already the newest version (16+257build1.1).
The following packages were automatically installed and are no longer required:
  libdrm-nouveau2 libdrm-radeon1 libgl1-amber-dri libglapi-mesa libllvm17t64 libxcb-dri2-0
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.
root@Training_svr:/# |

```

## Install PostgreSQL 18

```
# apt install -y postgresql-18 postgresql-client-18
```

```

root@Training_svr:/# apt install -y postgresql-18 postgresql-client-18
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
postgresql-18 is already the newest version (18.1-1.pgdg24.04+2).
postgresql-18 set to manually installed.
postgresql-client-18 is already the newest version (18.1-1.pgdg24.04+2).
postgresql-client-18 set to manually installed.
The following packages were automatically installed and are no longer required:
  libdrm-nouveau2 libdrm-radeon1 libgl1-amber-dri libglapi-mesa libllvm17t64 libxcb-dri2-0
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.
root@Training_svr:/#

```

## Install additional extensions

```
# apt install -y postgresql-18-pgvector
```

```

root@Training_svr:/# apt install -y postgresql-18-pgvector
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libdrm-nouveau2 libdrm-radeon1 libgl1-amber-dri libglapi-mesa libllvm17t64 libxcb-dri2-0
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  postgresql-18-pgvector
0 upgraded, 1 newly installed, 0 to remove and 2 not upgraded.
Need to get 259 kB of archives.
After this operation, 714 kB of additional disk space will be used.
Get:1 http://apt.postgresql.org/pub/repos/apt noble-pgdg/main amd64 postgresql-18-pgvector amd64 0.8.1-2.pgdg24.04+1 [259 kB]
Fetched 259 kB in 1s (176 kB/s)
Selecting previously unselected package postgresql-18-pgvector.
(Reading database ... 45403 files and directories currently installed.)
Preparing to unpack .../postgresql-18-pgvector_0.8.1-2.pgdg24.04+1_amd64.deb ...
Unpacking postgresql-18-pgvector (0.8.1-2.pgdg24.04+1) ...
Setting up postgresql-18-pgvector (0.8.1-2.pgdg24.04+1) ...
Processing triggers for postgresql-common (287.pgdg24.04+1) ...
Building PostgreSQL dictionaries from installed myspell/hunspell packages...
Removing obsolete dictionary files:
root@Training_svr:/# |

```

```
# apt install -y postgresql-18-postgis-3
```

```
root@Training_svr:/# apt install -y postgresql-18-postgis-3
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libdrm-nouveau2 libdrm-radeon1 libgl1-amber-dri libglapi-mesa libllvm17t64 libxcb-dri2-0
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  gdal-data gdal-plugins libaec0 libaom3 libarmadillo12 libarpack2t64 libblas3 libblosc1 libboost-serialization1.83.0
  libcfitsio10t64 libde265-0 libfreexl1 libfyba0t64 libgdal34t64 libgeos-c1t64 libgeos3.12.1t64 libgeotiff5 libgfortran5 libgif7
  libgmpxx4ldbl libhdf4-0-alt libhdf5-103-1t64 libhdf5-hl-100t64 libheif-plugin-aomdec libheif-plugin-aomenc
```

## Verify Installation

*Check PostgreSQL service status*

```
systemctl status postgresql
```

```
root@Training_svr:/# systemctl status postgresql
● postgresql.service - PostgreSQL RDBMS
   Loaded: loaded (/usr/lib/systemd/system/postgresql.service; enabled; preset: enabled)
   Active: active (exited) since Sun 2025-12-21 17:37:14 MSK; 1h 50min ago
   Main PID: 507 (code=exited, status=0/SUCCESS)

Dec 21 17:37:14 998-986-02 systemd[1]: Starting postgresql.service - PostgreSQL RDBMS...
Dec 21 17:37:14 998-986-02 systemd[1]: Finished postgresql.service - PostgreSQL RDBMS.
Dec 21 17:48:43 998-986-02 systemd[1]: Reloading postgresql.service - PostgreSQL RDBMS...
Dec 21 17:48:43 998-986-02 systemd[1]: Reloaded postgresql.service - PostgreSQL RDBMS.
root@Training_svr:/# |
```

*Check PostgreSQL versión*

```
psql -version
```

```
root@Training_svr:/# psql --version
psql (PostgreSQL) 18.1 (Ubuntu 18.1-1.pgdg24.04+2)
root@Training_svr:/#
```

List installed PostgreSQL packages

```
dpkg -l | grep postgresql
```

```

root@Training_svr:/# dpkg -l | grep postgresql
ii postgresql                    18+287.pgdg24.04+1      all      object-relational SQL database (supported
version)
ii postgresql-16                 16.11-1.pgdg24.04+1     amd64    The World's Most Advanced Open Source Rel
ational Database
ii postgresql-18                 18.1-1.pgdg24.04+2     amd64    The World's Most Advanced Open Source Rel
ational Database
ii postgresql-18-jit             18.1-1.pgdg24.04+2     amd64    LLVM JIT support for PostgreSQL 18
ii postgresql-18-pgvector        0.8.1-2.pgdg24.04+1     amd64    Open-source vector similarity search for
Postgres
ii postgresql-18-postgis-3       3.6.1+dfsg-1.pgdg24.04+1 amd64    Geographic objects support for PostgreSQL
18
ii postgresql-18-postgis-3-scripts 3.6.1+dfsg-1.pgdg24.04+1 all      Geographic objects support for PostgreSQL
18 -- SQL scripts
ii postgresql-client-16          16.11-1.pgdg24.04+1     amd64    front-end programs for PostgreSQL 16
ii postgresql-client-18          18.1-1.pgdg24.04+2     amd64    front-end programs for PostgreSQL 18
ii postgresql-client-common      287.pgdg24.04+1         all      manager for multiple PostgreSQL client ve
rsions
ii postgresql-common             287.pgdg24.04+1         all      PostgreSQL database-cluster manager
ii postgresql-contrib            16+257build1.1          all      additional facilities for PostgreSQL (sup
ported version)
root@Training_svr:/#

```

## PostgreSQL Initial Configuration

### Enable and Start Services

*Enable PostgreSQL to start on boot*

```
sudo systemctl enable postgresql
```

```

Synchronizing state of postgresql.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable postgresql
root@Training_svr:/#

```

### Check PostgreSQL Clusters

*List PostgreSQL clusters*

```
pg_lsclusters
```

```

root@Training_svr:/# pg_lsclusters
Ver Cluster Port Status Owner    Data directory          Log file
16  main     5433 down   postgres /var/lib/postgresql/16/main log/postgresql-%Y-%m-%d_%H%M%S.csv.csv
18  main     5432 online postgres /var/lib/postgresql/18/main log/postgresql-%Y-%m-%d_%H%M%S.csv.csv
root@Training_svr:/# |

```

## Configure PostgreSQL

### Set Password for PostgreSQL User

*Switch to postgres user*

```
sudo -i -u postgres
```

#### Chapter 2: PostgreSQL Installation and Initialization

```
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 5.15.167.4-microsoft-standard-WSL2 x86_64)
```

```
* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro
```

```
System information as of Sun Dec 21 19:34:42 MSK 2025
```

```
System load:  0.0      Processes:            47
Usage of /:   0.3% of 1006.85GB Users logged in:      2
Memory usage: 12%     IPv4 address for eth0: 192.168.248.237
Swap usage:   0%
```

```
This message is shown once a day. To disable it please create the
/var/lib/postgresql/.hushlogin file.
postgres@Trainingsvr:~$
```

*Access PostgreSQL shell*

**psql**

```
postgres@Trainingsvr:~$ psql
psql (18.1 (Ubuntu 18.1-1.pgdg24.04+2))
Type "help" for help.

postgres=#
```

*Inside psql, change postgres user password*

```
ALTER USER postgres PASSWORD 'postgres';
```

```
postgres=# ALTER USER postgres PASSWORD 'postgres';
ALTER ROLE
postgres=#
```

## Test PostgreSQL Connection

### *Local Connection Test*

Connect as postgres user

```
sudo -i -u postgres psql
```

```
root@Training_svr:/# sudo -i -u postgres psql
psql (18.1 (Ubuntu 18.1-1.pgdg24.04+2))
Type "help" for help.
```

```
postgres=# \l
```

| List of databases |          |          |                 |         |         |        |           |                         |
|-------------------|----------|----------|-----------------|---------|---------|--------|-----------|-------------------------|
| Name              | Owner    | Encoding | Locale Provider | Collate | Ctype   | Locale | ICU Rules | Access privileges       |
| postgres          | postgres | UTF8     | libc            | C.UTF-8 | C.UTF-8 |        |           |                         |
| template0         | postgres | UTF8     | libc            | C.UTF-8 | C.UTF-8 |        |           | =c/postgres +           |
| template1         | postgres | UTF8     | libc            | C.UTF-8 | C.UTF-8 |        |           | postgres=Ctc/postgres + |
| training          | postgres | UTF8     | libc            | C.UTF-8 | C.UTF-8 |        |           | =c/postgres +           |
| (4 rows)          |          |          |                 |         |         |        |           | postgres=Ctc/postgres   |

```
postgres=#
```

Run a test command

```
SELECT version();
```

```
postgres=# SELECT version();
```

| version                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------|
| PostgreSQL 18.1 (Ubuntu 18.1-1.pgdg24.04+2) on x86_64-pc-linux-gnu, compiled by gcc (Ubuntu 13.3.0-6ubuntu2~24.04) 13.3.0, 64-bit |

(1 row)

```
postgres=#
```

Create a test database

```
CREATE DATABASE training;
```

```
postgres=# CREATE DATABASE training;
ERROR: database "training" already exists
postgres=# \l
```

| List of databases |          |          |                 |         |         |        |           |                         |
|-------------------|----------|----------|-----------------|---------|---------|--------|-----------|-------------------------|
| Name              | Owner    | Encoding | Locale Provider | Collate | Ctype   | Locale | ICU Rules | Access privileges       |
| postgres          | postgres | UTF8     | libc            | C.UTF-8 | C.UTF-8 |        |           |                         |
| template0         | postgres | UTF8     | libc            | C.UTF-8 | C.UTF-8 |        |           | =c/postgres +           |
| template1         | postgres | UTF8     | libc            | C.UTF-8 | C.UTF-8 |        |           | postgres=Ctc/postgres + |
| training          | postgres | UTF8     | libc            | C.UTF-8 | C.UTF-8 |        |           | =c/postgres +           |
| (4 rows)          |          |          |                 |         |         |        |           | postgres=Ctc/postgres   |

```
postgres=#
```

List databases

```
\l
```

```
postgres=# \l
```

| List of databases |          |          |        |          |         |         |        |           |                       |
|-------------------|----------|----------|--------|----------|---------|---------|--------|-----------|-----------------------|
| Name              | Owner    | Encoding | Locale | Provider | Collate | Ctype   | Locale | ICU Rules | Access privileges     |
| postgres          | postgres | UTF8     | libc   |          | C.UTF-8 | C.UTF-8 |        |           |                       |
| template0         | postgres | UTF8     | libc   |          | C.UTF-8 | C.UTF-8 |        |           | =c/postgres +         |
| template1         | postgres | UTF8     | libc   |          | C.UTF-8 | C.UTF-8 |        |           | =c/postgres +         |
| training          | postgres | UTF8     | libc   |          | C.UTF-8 | C.UTF-8 |        |           | postgres=Ctc/postgres |

```
(4 rows)

postgres=#
```

## Create a New User and Database

Create a new user

```
sudo -u postgres createuser --interactive -pwprompt
```

```
postgres@Trainingsvr:~$ exit
logout
root@Training_svr:/# sudo -u postgres createuser --interactive -pwprompt
Shall the new role be a superuser? (y/n) y
root@Training_svr:/#
```

Create database owned by the new user

```
sudo -u postgres createdb -O postgres demo
```

```
root@Training_svr:/# sudo -u postgres createdb -O postgres demo
root@Training_svr:/# |
```

Grant privileges

```
sudo -u postgres psql -c "GRANT ALL PRIVILEGES ON DATABASE demo TO postgres;"
```

```
root@Training_svr:/# sudo -u postgres psql -c "GRANT ALL PRIVILEGES ON DATABASE demo TO postgres;"
GRANT
root@Training_svr:/# |
```

## Configure Remote Access

### Edit PostgreSQL Configuration Files

Backup original config files for PostgreSQL 18

```
sudo cp /etc/postgresql/18/main/postgresql.conf /etc/postgresql/18/main/postgresql.conf.backup
sudo cp /etc/postgresql/18/main/pg_hba.conf /etc/postgresql/18/main/pg_hba.conf.backup
```

## Chapter 2: PostgreSQL Installation and Initialization

```

root@Training_svr:/# ls -lrt /etc/postgresql/18/main/*
-rw-r--r-- 1 postgres postgres 315 Dec 21 19:13 /etc/postgresql/18/main/environment
-rw-r--r-- 1 postgres postgres 317 Dec 21 19:13 /etc/postgresql/18/main/start.conf
-rw-r----- 1 postgres postgres 2640 Dec 21 19:13 /etc/postgresql/18/main/pg_ident.conf
-rw-r----- 1 postgres postgres 5924 Dec 21 19:13 /etc/postgresql/18/main/pg_hba.conf
-rw-r--r-- 1 postgres postgres 143 Dec 21 19:13 /etc/postgresql/18/main/pg_ctl.conf
-rw-r--r-- 1 postgres postgres 30374 Dec 21 19:13 /etc/postgresql/18/main/postgresql.conf

/etc/postgresql/18/main/conf.d:
total 0
root@Training_svr:/# sudo cp /etc/postgresql/18/main/postgresql.conf /etc/postgresql/18/main/postgresql.conf.backup
sudo cp /etc/postgresql/18/main/pg_hba.conf /etc/postgresql/18/main/pg_hba.conf.backup
root@Training_svr:/# ls -lrt /etc/postgresql/18/main/*
-rw-r--r-- 1 postgres postgres 315 Dec 21 19:13 /etc/postgresql/18/main/environment
-rw-r--r-- 1 postgres postgres 317 Dec 21 19:13 /etc/postgresql/18/main/start.conf
-rw-r----- 1 postgres postgres 2640 Dec 21 19:13 /etc/postgresql/18/main/pg_ident.conf
-rw-r----- 1 postgres postgres 5924 Dec 21 19:13 /etc/postgresql/18/main/pg_hba.conf
-rw-r--r-- 1 postgres postgres 143 Dec 21 19:13 /etc/postgresql/18/main/pg_ctl.conf
-rw-r--r-- 1 postgres postgres 30374 Dec 21 19:13 /etc/postgresql/18/main/postgresql.conf
-rw-r--r-- 1 root root 30374 Dec 21 20:54 /etc/postgresql/18/main/postgresql.conf.backup
-rw-r----- 1 root root 5924 Dec 21 20:54 /etc/postgresql/18/main/pg_hba.conf.backup

/etc/postgresql/18/main/conf.d:
total 0
root@Training_svr:/# |

```

## Configure postgresql.conf for PostgreSQL 18

*Edit the main configuration file for PostgreSQL 18*

```
vi /etc/postgresql/18/main/postgresql.conf
```

```

data_directory = '/var/lib/postgresql/18/main'          # use data in another directory
                                                         # (change requires restart)
hba_file = '/etc/postgresql/18/main/pg_hba.conf'       # host-based authentication file
                                                         # (change requires restart)
ident_file = '/etc/postgresql/18/main/pg_ident.conf'   # ident configuration file
                                                         # (change requires restart)

# If external_pid_file is not explicitly set, no extra PID file is written.
external_pid_file = '/var/run/postgresql/18-main.pid'  # write an extra PID file
                                                         # (change requires restart)

#-----
# CONNECTIONS AND AUTHENTICATION
#-----

# - Connection Settings -

#listen_addresses = 'localhost'          # what IP address(es) to listen on;
#                                         # comma-separated list of addresses;
#                                         # defaults to 'localhost'; use '*' for all
#                                         # (change requires restart)
port = 5432                              # (change requires restart)
max_connections = 100                    # (change requires restart)
#reserved_connections = 0                 # (change requires restart)
#superuser_reserved_connections = 3       # (change requires restart)
unix_socket_directories = '/var/run/postgresql' # comma-separated list of directories
#                                         # (change requires restart)
#unix_socket_group = ''                  # (change requires restart)
#unix_socket_permissions = 0777          # begin with 0 to use octal notation
#                                         # (change requires restart)
#bonjour = off                           # advertise server via Bonjour

```

## Configure pg\_hba.conf for PostgreSQL 18

*Edit the host-based authentication file*

```
vi /etc/postgresql/18/main/pg_hba.conf
```

```

# Database administrative login by Unix domain socket
local  all             postgres                                peer

# TYPE  DATABASE  USER  ADDRESS  METHOD

# "local" is for Unix domain socket connections only
local  all             all                                     peer
# IPv4 local connections:
host   all             all             127.0.0.1/32          scram-sha-256
# IPv6 local connections:
host   all             all             ::1/128               scram-sha-256
# Allow replication connections from localhost, by a user with the
# replication privilege.
local  replication     all                                     peer
host   replication     all             127.0.0.1/32          scram-sha-256
host   replication     all             ::1/128               scram-sha-256

```

## Restart PostgreSQL 18

### *Restart PostgreSQL service*

```
systemctl restart postgresql@18-main
```

```
root@Training_svr:/# systemctl status postgresql@18-main
● postgresql@18-main.service - PostgreSQL Cluster 18-main
   Loaded: loaded (/usr/lib/systemd/system/postgresql@.service; enabled-runtime; preset: enabled)
   Active: active (running) since Sun 2025-12-21 21:16:48 MSK; 15s ago
     Process: 13441 ExecStart=/usr/bin/pg_ctlcluster --skip-systemctl-redirect 18-main start (code=exited, status=0/SUCCESS)
    Main PID: 13446 (postgres)
      Tasks: 11 (limit: 4597)
     Memory: 28.7M ()
    CGroup: /system.slice/system-postgresql.slice/postgresql@18-main.service
            └─13446 /usr/lib/postgresql/18/bin/postgres -D /var/lib/postgresql/18/main -c config_file=/etc/postgresql/18/main/postg
            └─13447 "postgres: 18/main: logger "
            └─13448 "postgres: 18/main: io worker 0"
            └─13449 "postgres: 18/main: io worker 1"
            └─13450 "postgres: 18/main: io worker 2"
            └─13451 "postgres: 18/main: checkpointer "
            └─13452 "postgres: 18/main: background writer "
            └─13454 "postgres: 18/main: walwriter "
            └─13455 "postgres: 18/main: autovacuum launcher "
            └─13456 "postgres: 18/main: archiver failed on 00000001000000000000000001"
            └─13457 "postgres: 18/main: logical replication launcher "

Dec 21 21:16:45 Trainingsvr systemd[1]: Starting postgresql@18-main.service - PostgreSQL Cluster 18-main...
Dec 21 21:16:48 Trainingsvr systemd[1]: Started postgresql@18-main.service - PostgreSQL Cluster 18-main.
lines 1-22/22 (END)
```

## Configure Firewall

*Allow PostgreSQL port (5432) through firewall*

```
ufw allow 5432/tcp
```

```
root@Training_svr:/# ufw allow 5432/tcp
Rule added
Rule added (v6)
root@Training_svr:/# |
```

R

## Install Additional Tools for PostgreSQL 18

### Install Command-line Tools

*# Install useful PostgreSQL tools*

```
sudo apt install -y pgtop pgcli
```

```

root@Training_svr:/# sudo apt install -y pgtop pgcli
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libdrm-nouveau2 libdrm-radeon1 libgl1-amber-dri libglapi-mesa libllvm17t64 libxcb-dri2-0
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  python3-cli-helpers python3-dateutil python3-pendulum python3-pgspecial python3-prompt-toolkit python3-psycopg python3-pytzdata
  python3-setproctitle python3-sqlparse python3-tabulate python3-terminaltables python3-wcwidth
Suggested packages:
  python3-psycopg-pool python-psycopg-doc python-sqlparse-doc python3-terminaltables-doc
The following NEW packages will be installed:
  pgcli pgtop python3-cli-helpers python3-dateutil python3-pendulum python3-pgspecial python3-prompt-toolkit python3-psycopg
  python3-pytzdata python3-setproctitle python3-sqlparse python3-tabulate python3-terminaltables python3-wcwidth
0 upgraded, 14 newly installed, 0 to remove and 2 not upgraded.
Need to get 815 kB of archives.
After this operation, 4892 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 python3-tabulate all 0.8.10-1 [38.2 kB]
Get:2 http://apt.postgresql.org/pub/repos/apt noble-pgdg/main amd64 pgtop amd64 4.1.3-2.pgdg24.04+1 [43.1 kB]

```

*# Install backup tools*

```
sudo apt install -y postgresql-client-common pgbackrest
```

```

root@Training_svr:/# sudo apt install -y postgresql-client-common pgbackrest
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
postgresql-client-common is already the newest version (287.pgdg24.04+1).
postgresql-client-common set to manually installed.
The following packages were automatically installed and are no longer required:
  libdrm-nouveau2 libdrm-radeon1 libgl1-amber-dri libglapi-mesa libllvm17t64 libxcb-dri2-0
Use 'sudo apt autoremove' to remove them.
Suggested packages:
  pgbackrest-doc check-pgbackrest
The following NEW packages will be installed:
  libssh2-1t64 pgbackrest
0 upgraded, 2 newly installed, 0 to remove and 2 not upgraded.
Need to get 679 kB of archives.
After this operation, 1879 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 libssh2-1t64 amd64 1.11.0-4.1build2 [120 kB]
Get:2 http://apt.postgresql.org/pub/repos/apt noble-pgdg/main amd64 pgbackrest amd64 2.57.0-1.pgdg24.04+1 [560 kB]
Fetched 679 kB in 1s (661 kB/s)
Selecting previously unselected package libssh2-1t64:amd64.

```

## Database Operations

*# Backup a database*

```
sudo -u postgres pg_dump training > training_backup.sql
```

```

-rw-r--r--  1 root root  19107 Dec 21 22:00 training_backup.sql
root@Training_svr:/# |

```

*# Restore a database*

```
sudo -u postgres psql -d training -f training_backup.sql
```

```

root@Training_svr:/# sudo -u postgres psql -d training -f training_backup.sql
SET
SET
SET
SET
SET
SET
SET
  set_config
-----
(1 row)

SET
SET
SET
SET
CREATE EXTENSION
COMMENT

```

*Create backup with custom format*

```
sudo -u postgres pg_dump -Fc training > training_backup.dump
```

*Restore custom format backup*

```
sudo -u postgres pg_restore -d training training_backup.dump
```

## Verify Complete Installation

*Run comprehensive test*

```

echo "=== PostgreSQL Installation Test ==="
echo "1. Service Status:" && sudo systemctl is-active postgresql
echo "2. Version:" && psql --version
echo "3. Cluster Status:" && pg_lsclusters
echo "4. Connection Test:" && sudo -u postgres psql -c "SELECT version();"
echo "5. Database List:" && sudo -u postgres psql -c "\l"

```

```
root@training_svr:/# echo "=== PostgreSQL Installation Test ==="
echo "1. Service Status:" && sudo systemctl is-active postgresql
echo "2. Version:" && psql --version
echo "3. Cluster Status:" && pg_lsclusters
echo "4. Connection Test:" && sudo -u postgres psql -c "SELECT version();"
echo "5. Database List:" && sudo -u postgres psql -c "\l"
=== PostgreSQL Installation Test ===
1. Service Status:
active
2. Version:
psql (PostgreSQL) 18.1 (Ubuntu 18.1-1.pgdg24.04+2)
3. Cluster Status:
Ver Cluster Port Status Owner    Data directory          Log file
16 main     5433 down   postgres /var/lib/postgresql/16/main log/postgresql-%Y-%m-%d_%H%M%S.csv.csv
18 main     5432 online postgres /var/lib/postgresql/18/main log/postgresql-%Y-%m-%d_%H%M%S.csv.csv
4. Connection Test:
                                version
-----
PostgreSQL 18.1 (Ubuntu 18.1-1.pgdg24.04+2) on x86_64-pc-linux-gnu, compiled by gcc (Ubuntu 13.3.0-6ubuntu2~24.04) 13.3.0, 64-bit
(1 row)

5. Database List:
                                List of databases
 Name | Owner  | Encoding | Locale Provider | Collate | Ctype | Locale | ICU Rules | Access privileges
-----+-----+-----+-----+-----+-----+-----+-----+-----
demo  | postgres | UTF8     | libc            | C.UTF-8 | C.UTF-8 |         |            | =Tc/postgres +
      |          |          |                  |          |          |         |            | postgres=Ctc/postgres
postgres | postgres | UTF8     | libc            | C.UTF-8 | C.UTF-8 |         |            | =c/postgres +
      |          |          |                  |          |          |         |            | postgres=Ctc/postgres
template0 | postgres | UTF8     | libc            | C.UTF-8 | C.UTF-8 |         |            | =c/postgres +
      |          |          |                  |          |          |         |            | postgres=Ctc/postgres
template1 | postgres | UTF8     | libc            | C.UTF-8 | C.UTF-8 |         |            | =c/postgres +
      |          |          |                  |          |          |         |            | postgres=Ctc/postgres
training | postgres | UTF8     | libc            | C.UTF-8 | C.UTF-8 |         |            |
```